

The processes of spermatogenesis and oogenesis in humans are similar in that they both:

- ☐ A. begin at puberty. [7%]
- ✓ ☒ B. involve cells that undergo more than one round of cell division. [83%]
- ☐ C. ultimately result in daughter cells that are all approximately equal in size. [5%]
- ☐ D. occur throughout the life span of the organism. [3%]

Correct

83% answered correctly

Explanation:

	Spermatogenesis (males ♂)	Oogenesis (females ♀)
Start of process	Puberty	Prior to birth (in fetal stage)
End of process	Never, spermatogenesis continues throughout life	Stops at menopause (cessation of menstruation)
Gonads (site of production)	Testes	Ovaries
Timing of gamete production & release	Continuous	Monthly
Meiotic cell divisions	Occur without interruption & produce similarly sized daughter cells after each round	Arrested at certain stages & produce cells of unequal size
Total gametes produced in a single cycle	4 haploid sperm cells	1 haploid ovum plus 2-3 polar bodies that eventually degenerate

In humans, the generation of reproductive cells (gametes) occurs in reproductive glands known as the gonads. Before puberty in males, stem cells called spermatogonia undergo continuous **mitotic divisions** within the male gonads (ie, **testes**) to yield identical daughter cells. At **puberty**, spermatogenesis (sperm production) begins and continues throughout the male's life.

In **spermatogenesis**, some of the daughter cells produced from mitotic divisions of spermatogonia become primary spermatocytes. **Primary spermatocytes** then become mature sperm via **meiosis**, a process involving **two rounds of cell division known as meiosis I and II**. A primary spermatocyte that undergoes meiosis I yields two *identical* haploid cells labeled secondary spermatocytes. Each secondary spermatocyte then undergoes meiosis II to produce four haploid spermatids that are approximately equal in size to each other and develop into mature **sperm**.

In contrast to spermatogenesis, **oogenesis** (oocyte production) begins before birth, *not* at puberty, in the female gonads (ie, **ovaries**) (**Choice A**). In the female embryo, oogonia (stem cells) undergo mitosis to produce primary oocytes. Each primary oocyte is surrounded by a saclike structure called a follicle. Like primary spermatocytes, **primary oocytes** must **also undergo meiosis** to mature, although these oocytes become arrested at prophase I of meiosis I until puberty. At puberty, one primary oocyte is selected during each **menstrual cycle** to complete meiosis I.

Unlike in spermatogenesis, meiosis I in oogenesis yields two haploid cells of *unequal* size, the *larger* one being a secondary oocyte and the *smaller* one being a polar body that eventually degenerates (**Choice C**). The secondary oocyte then begins meiosis II but is arrested at metaphase II. The secondary oocyte completes meiosis II and fully matures only if **fertilization** occurs. To be fertilized, the secondary oocyte must be released during the ovulation phase of the menstrual cycle. During this phase, the follicle ruptures from the ovary and the oocyte enters the **fallopian tube** to be fertilized by mature sperm.

(Choice D) Oogenesis begins before birth but ceases in older women when ovarian production of female sex hormones declines (ie, menopause). In contrast, spermatogenesis begins at puberty and continues throughout the male's life. Neither process occurs continuously throughout an organism's entire life span.

Educational objective:

Both spermatogenesis and oogenesis involve cells that undergo meiosis I and II. However, oogenesis in females begins in the female embryo and ends at menopause, whereas spermatogenesis in males does not begin until puberty and continues throughout a male's life.

Time spent:0 sec

Subject:
Biology

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System:
Reproduction